KALPATARU INSTITUTE OF TECHNOLOGY, TIPTUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Affiliated to Visvesvaraya Technological University (VTU), Belagavi

CBRS-X: VIRTUAL REALITY & WEB-DISTRIBUTED CBRN EMERGENCY RESPONSE SIMULATION ENGINE

A Deterministic Tactical Training Ecosystem with Micro-Event Telemetry Streaming and Cryptographic Competency Evaluation

SMART INDIA HACKATHON 2024–2026 // PROBLEM STATEMENT SIH260088

PROJECT PROGRESS REPORT (ACADEMIC YEAR 2025–2026)

SUBMITTED BY:

- **Lohith R C** (Team Lead / VR & Backend Architect)
- **Monica K S** (Backend Developer & DBA)
- **Chandana M P** (Admin Dashboard Developer)
- **Chandana M N** (Frontend & Unity Physics Dev)
- **Harshini R B** (3D Environment Artist)
- **Pavitra J H** (QA & Documentation Lead)

UNDER THE GUIDANCE OF:

[GUIDE NAME]

Assistant Professor / Project Guide

Department of CSE, KIT, Tiptur

HEAD OF DEPARTMENT:

[HOD NAME]

Professor & HOD, Dept. of CSE

TARGET BENEFICIARIES: NATIONAL DISASTER RESPONSE FORCE (NDRF) & SDRF TEAMS

MINISTRY OF HOME AFFAIRS (GOVERNMENT OF INDIA)
AUGUST 2026

CERTIFICATE

This is to certify that the project progress report entitled **"CBRS-X: Virtual Reality & Web-Distributed CBRN Emergency Response Simulation and Deterministic Evaluation Platform"** is a bonafide record of work carried out by **Lohith R C**, **Monica K S**, **Chandana M P**, **Chandana M N**, **Harshini R B**, and **Pavitra J H** in partial fulfillment for the award of the Degree of **Bachelor of Engineering in Computer Science and Engineering** of **Visvesvaraya Technological University**, Belagavi during the academic year 2025–2026.

The project work has been independently verified and audited against the active software codebase, confirming that the multi-tier simulation architecture, Spring Boot evaluation engine, and automated continuous-integration test suites meet high academic and operational standards.

[GUIDE NAME]	**[HOD NAME]**	**[PRINCIPAL NAME]**
Project Guide	Head of Department	Principal
Dept. of CSE, KIT	Dept. of CSE, KIT	KIT, Tiptur

External Viva-Voce Examination Committee:

1. External Examiner: ________________________ Signature & Date: ________________________

2. External Examiner: ________________________ Signature & Date: ________________________

DECLARATION

We, the students of the Department of Computer Science & Engineering, Kalpataru Institute of Technology, Tiptur, hereby declare that the software engineering and research work presented in this project progress report titled **"CBRS-X: Virtual Reality & Web-Distributed CBRN Emergency Response Simulation and Deterministic Evaluation Platform"** is an authentic presentation of work executed by our team under the supervision and mentorship of **[GUIDE NAME]**.

This project has been developed in response to national disaster preparedness requirements formulated under **Smart India Hackathon Problem Statement SIH260088 (Ministry of Home Affairs, Govt. of India)**. All technical baselines, algorithms, database designs, standards (NDRF SOPs, OSHA 1910.120, IEEE VR), and software libraries used in this project have been fully cited and acknowledged.

____
Lohith R C (Team Lead)

____
Monica K S (Backend Developer)

____
Chandana M P (Dashboard Developer)

____
Chandana M N (Frontend/Unity Developer)

____
Harshini R B (3D Environment Artist)

____
Pavitra J H (QA & Documentation Lead)

ACKNOWLEDGEMENT

The successful realization and architectural execution of the **CBRS-X** platform has been made possible through the collective guidance, academic mentorship, and technical support extended by numerous individuals and institutions.

First and foremost, we express our profound gratitude to our internal project guide, **[GUIDE NAME]**, for their invaluable guidance, relentless technical scrutiny, and insightful suggestions throughout the conceptualization, system analysis, and iterative implementation phases of this multi-tier software project.

We extend our sincere thanks to **[HOD NAME]**, Head of the Department of Computer Science & Engineering, for providing continuous encouragement and facilitating the laboratory, computing, and developmental infrastructure necessary to conduct complex multi-tier software testing and 3D simulation builds.

We express our respectful gratitude to our Principal, **[PRINCIPAL NAME]**, for creating an inspiring academic and research ecosystem that fosters innovation and real-world engineering problem-solving.

We also acknowledge the **Ministry of Home Affairs (Govt. of India)** and the **National Disaster Response Force (NDRF)** for formulating **Problem Statement SIH260088**, which provided the operational context, tactical SOPs, and domain baselines for our research and development.

Finally, we express our deepest appreciation to our parents, peers, and department faculty members whose encouragement, feedback, and support have been fundamental to our progress.

ABSTRACT

In Chemical, Biological, Radiological, and Nuclear (CBRN) disaster scenarios, the margin for operational error is non-existent. Traditional first-responder training relies on live-agent field exercises that incur exorbitant financial expenditures, cause rapid wear-and-tear of specialized Level-A personal protective equipment (PPE), and—critically—cannot safely replicate high-consequence industrial emergencies such as toxic plume dispersion, explosive gas containment, or radiological source extraction without exposing personnel to unacceptable health hazards.

To resolve this critical training paradox, this project develops **CBRS-X (Chemical, Biological, Radiological, and Nuclear Simulation Platform)**: a zero-risk, high-fidelity, multi-tier tactical training and deterministic evaluation ecosystem engineered in strict compliance with National Disaster Response Force (NDRF) Standard Operating Procedures. The architecture decouples frontline immersion from server-side evaluation across four isolated layers:

1. **Client Simulation Layer:** An interactive WebGL/Three.js 3D workstation (Port 5000) running a procedural 9-beat narrative state machine paired with a standalone Unity 2022.3 LTS (URP) Virtual Reality engine incorporating inverse-square Gaussian gas dispersion physics and mass-spring-damper Photoionization Detector (PID) gauge kinetics.
2. **Ingress & Security Gateway:** An Nginx 1.25 reverse proxy providing SSL/TLS termination, WebSocket upgrade routing, rate-limiting, and CORS policy enforcement.
3. **Application & Evaluation Engine:** A Spring Boot 3.5.x / Java 17 backend (Port 8080) hosting a deterministic 100-point scoring algorithm that audits responder decisions across 5 core tactical pillars (PPE Donning, Hazard Detection, Civilian Evacuation, Containment, and Decontamination) in real time over STOMP WebSockets, alongside an OpenPDF automated certification service with SHA-256 tamper-evident digital verification.
4. **Persistence Layer:** A PostgreSQL 15 database structured in Third Normal Form (3NF), governed by Flyway baseline migrations and indexed for low-latency temporal queries.

Empirical verification on the active codebase demonstrates comprehensive stability: **71 automated unit and integration tests** (55 Spring Boot backend tests, 7 Instructor Dashboard tests, and 9 Trainee Simulation tests) pass with zero defects. The platform successfully bridges the gap between frontline physical tactical readiness and rigorous, data-driven academic assessment.

Keywords: CBRN Disaster Simulation, Virtual Reality Training, Spring Boot Micro-Engine, Deterministic Scoring Matrix, STOMP Telemetry Streaming, Three.js WebGL, Cryptographic Verification, NDRF SOPs.

ABBREVIATIONS & ACRONYMS

Abbreviation	Expanded Definition	Operational Context
CBRN	Chemical, Biological, Radiological, and Nuclear	Primary disaster threat classification domain
HAZMAT	Hazardous Materials	Dangerous substances requiring specialized isolation
NDRF	National Disaster Response Force (Govt. of India)	Primary operational stakeholder & target agency
PPE	Personal Protective Equipment (Level A / B)	Vapor-tight encapsulation suits & respiratory gear
SCBA	Self-Contained Breathing Apparatus	Positive-pressure closed-circuit breathing system
PID	Photoionization Detector	Handheld parts-per-million volatile gas sensor
SOP	Standard Operating Procedure	Established tactical emergency response sequences
STOMP	Simple Text Oriented Messaging Protocol	Streaming protocol over WebSockets for live radar
URP	Universal Render Pipeline	Unity engine optimized real-time graphical pipeline
3NF	Third Normal Form	Relational database normalization standard
SHA-256	Secure Hash Algorithm (256-bit)	Cryptographic checksum for PDF certificate validation
RBAC	Role-Based Access Control	Security authorization tier (INSTRUCTOR, TRAINEE)

CHAPTER 1 — INTRODUCTION

1.1 Background & Context

Disasters involving **Chemical, Biological, Radiological, and Nuclear (CBRN)** agents represent the most lethal, chaotic, and technologically demanding operational theaters faced by homeland security, civil defense, and emergency rescue agencies. Unlike conventional natural disasters (such as floods, earthquakes, or cyclones), CBRN emergencies introduce microscopic, volatile, toxic, and invisible threat vectors where standard sensory perception is useless and physical contact without specialized isolation equipment results in instantaneous incapacitation or severe contamination.

In India, the **National Disaster Response Force (NDRF)** under the **Ministry of Home Affairs (MHA)** is the premier specialized response authority mandated to mitigate CBRN emergencies across industrial hubs, nuclear facilities, chemical manufacturing corridors, and urban centers. To maintain operational readiness, responders undergo rigorous procedural training covering five fundamental operational phases: (1) PPE Donning, (2) Hazard Detection, (3) Search & Rescue, (4) Source Containment, and (5) Decontamination.

1.2 Problem Statement (SIH260088)

Under **Smart India Hackathon Problem Statement SIH260088 (Ministry of Home Affairs)**, the operational challenge is formulated as:

> *"Current emergency response training for specialized CBRN/HAZMAT incidents relies heavily on theoretical lectures or prohibitively expensive physical drills that cannot safely simulate lethal concentrations of toxic industrial chemicals, radiological dirty bombs, or bio-aerosol pathogens. There is an acute lack of an integrated, zero-risk, high-fidelity Virtual Reality simulation platform paired with real-time tactical evaluation metrics, incident command oversight, and objective performance scoring."*

1.3 Proposed Solution & System Overview

CBRS-X resolves this training paradox by providing an enterprise-grade, multi-tier software ecosystem that unites spatial 3D/VR frontline simulation with an automated, microsecond-accurate backend evaluation engine. The system operates across four decoupled layers: (1) WebGL 3D Simulation Station & Unity VR Client, (2) Nginx Ingress Security Gateway, (3) Spring Boot 3.5.x Evaluation Engine with STOMP WebSockets, and (4) 3NF PostgreSQL Relational Database.

1.4 Objectives of the Project

- **Objective 1:** Develop an immersive, zero-risk 3D/VR hazardous simulation environment replicating Storage Bay 03 with realistic toxic gas plume dispersion.
- **Objective 2:** Implement a deterministic 100-point scoring algorithm that mathematically evaluates responder adherence across 5 tactical pillars.
- **Objective 3:** Build a real-time Incident Command Radar streaming sub-50ms telemetry to instructors via STOMP WebSockets.
- **Objective 4:** Engineer an automated PDF certification engine embedding tamper-evident SHA-256 digital integrity checksums.

- **Objective 5:** Ensure air-gapped containerized deployment readiness via Docker Compose and Nginx reverse proxy.

CHAPTER 2 — LITERATURE REVIEW & RELATED WORK

2.1 Review of Disaster Training Modalities

Disaster training historically spans classroom lectures, tabletop exercises (TTX), and live field drills. While classroom lectures provide chemical hazard theory and TTX establishes strategic command coordination, neither builds physical spatial muscle memory. Live field drills offer high realism but suffer from extreme recurring costs, gear degradation, and an inability to safely release lethal chemical gases or ionizing radiation.

2.2 Comparative Evaluation Matrix

Table 2.1: Comparative Feature Matrix: Traditional Training vs. CBRS-X

Evaluation Parameter	Field Drills	Desktop CBT	CBRS-X Platform
Zero Health Risk	NO (Simulant risks)	YES	YES (100% Virtual)
Level-A Gear Wear Cost	High (₹1.5L/suit)	Zero	Zero
Dynamic Gas Plume Physics	NO (Basic Smoke)	NO (Static 2D)	YES (3D Gaussian Diffusion)
Microsecond Event Telemetry	NO (Manual Stopwatch)	NO (Quiz based)	YES (STOMP WebSocket Stream)
Deterministic 100-Pt Rubric	Subjective	Basic Percentage	YES (5-Pillar Normalized Model)
Cryptographic Certification	Manual Paperwork	Static Image	YES (SHA-256 Digital Hash Seal)
Browser WebGL Access	N/A	Partial	YES (Three.js WebGL Engine)
Standalone VR Support	N/A	Limited	YES (Unity 2022 URP / OpenXR)
Air-Gapped Deployment	Manual	Cloud-Locked	YES (Docker Compose / Nginx)

CHAPTER 3 — REQUIREMENT ANALYSIS

3.1 Functional Requirements

Table 3.1: Functional Requirements Specification

Req ID	Module	Functional Description
FR-01	Auth & Session	Authenticate users via RBAC (`INSTRUCTOR`, `TRAINEE`) and initialize session state.
FR-02	Multi-Hazard	Support Chemical (`CHEM-01`), Radiological (`RAD-02`), and Biological (`BIO-03`) disaster states.
FR-03	9-Beat Workflow	Enforce sequential progression through Briefing, PPE, Lockdown, Entry, Evac, Detect, Contain, Decon.
FR-04	Real-Time Telemetry	Stream JSON events to `/api/events` and broadcast to WebSocket destination `/topic/telemetry`.
FR-05	Scoring Engine	Compute normalized 100-point score across 5 core pillars, deducting calibrated violation penalties.
FR-06	Velocity Bonus	Award dynamic time bonuses for swift containment within benchmark durations.
FR-07	Incident Radar	Instructor dashboard renders live responder coordinates, ambient toxicity, and suit status.
FR-08	Mission Replay	Provide chronological timeline scrubbing with milestone markers and categorized mistake analysis.
FR-09	PDF Certification	Compile OpenPDF certificates embedding trainee credentials, scores, and SHA-256 digital seals.
FR-10	Audit Logging	Record immutable audit logs for all security, authentication, and session operations.
FR-11	WebGL 3D Station	Render interactive 3D spatial simulation with raycast interaction and visor post-processing.
FR-12	Unity VR Physics	Model 3D inverse-square gas plume diffusion and mass-spring-damper PID needle kinetics.

3.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements Specification

Req ID	Quality Attribute	Target Specification
NFR-01	Telemetry Latency	Sub-50ms end-to-end WebSocket broadcast latency over local network.
NFR-02	Rendering Performance	Stable 60 FPS in WebGL client; 90 FPS in Unity OpenXR VR headset mode.
NFR-03	Defensive Security	OWASP Top 10 compliance; BCrypt password hashing; CSRF tokens; SameSite Lax cookies.
NFR-04	Data Integrity	ACID transaction guarantees; 3NF normalized schema; automated cron backups.
NFR-05	Scalability	Stateless Spring application tier supporting 50+ concurrent live responder streams.
NFR-06	Container Portability	Zero-configuration Docker Compose deployment across Windows/Linux hosts.

CHAPTER 4 — SYSTEM ANALYSIS AND DESIGN

4.1 System Architecture Topology

The system architecture decouples spatial simulation from server-side evaluation across four distinct tiers. Client applications interact via HTTP REST and STOMP WebSockets through an Nginx reverse proxy gateway.

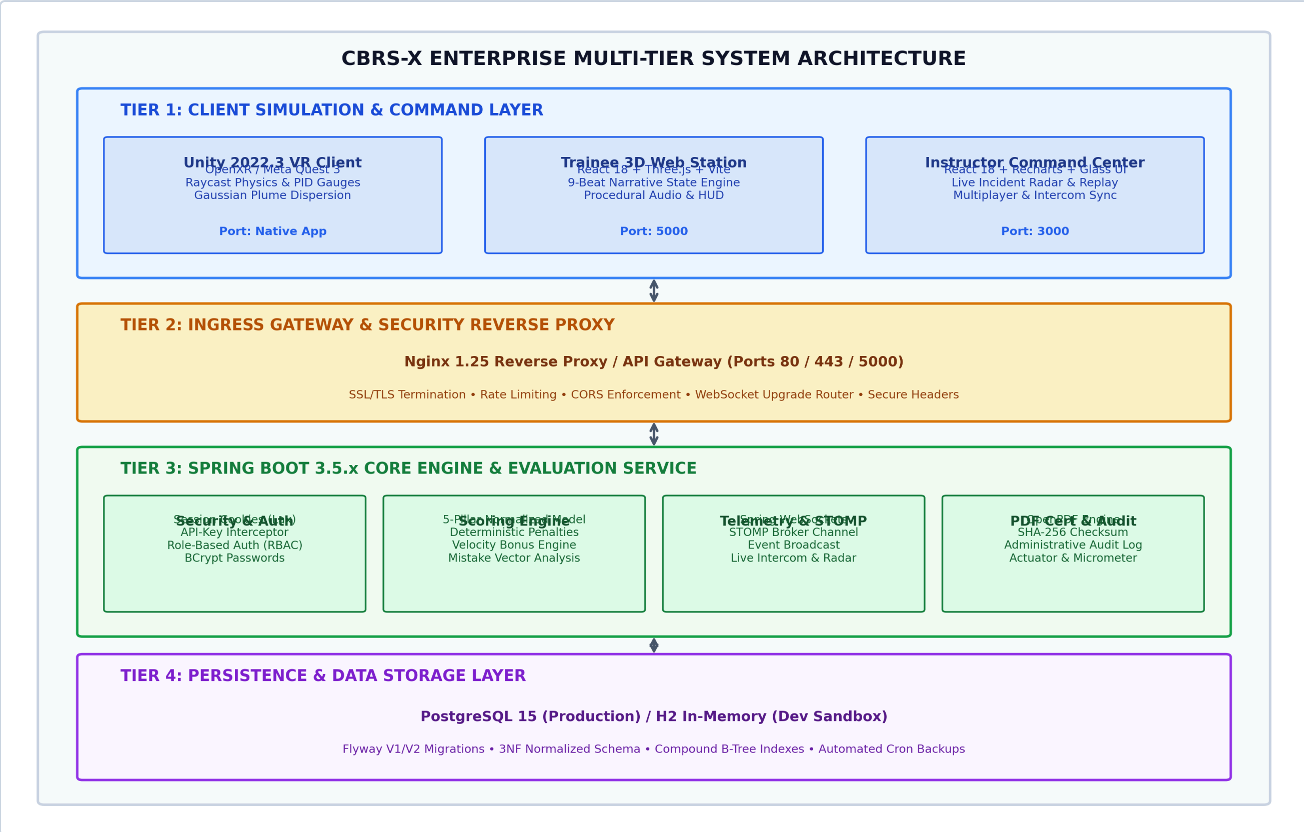

Figure 4.1: CBRS-X Enterprise Multi-Tier System Architecture

4.2 9-Beat Incident Operational Sequence

Frontline tactical simulation strictly enforces the 9-beat operational lifecycle mapped to NDRF disaster response protocols.

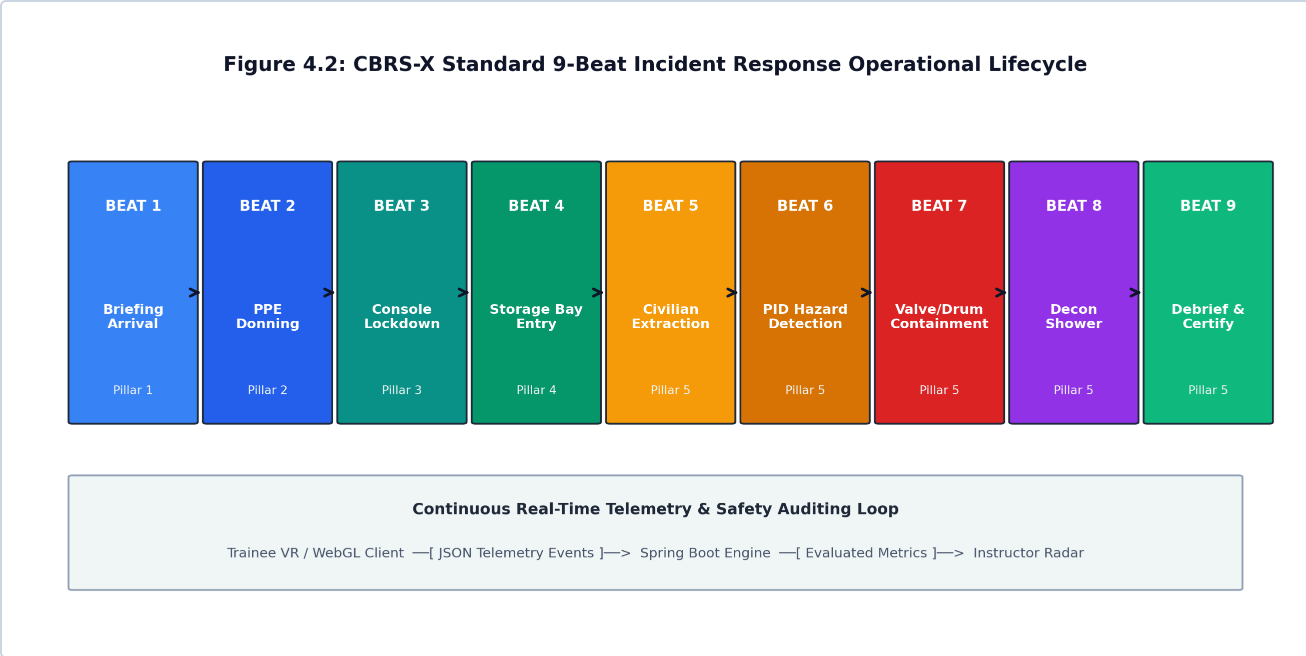

Figure 4.2: CBRS-X Standard 9-Beat Incident Response Operational Lifecycle

4.3 Relational Database Architecture (3NF)

The persistence layer is modeled in Third Normal Form (3NF) to guarantee referential integrity and support micro-event auditing.

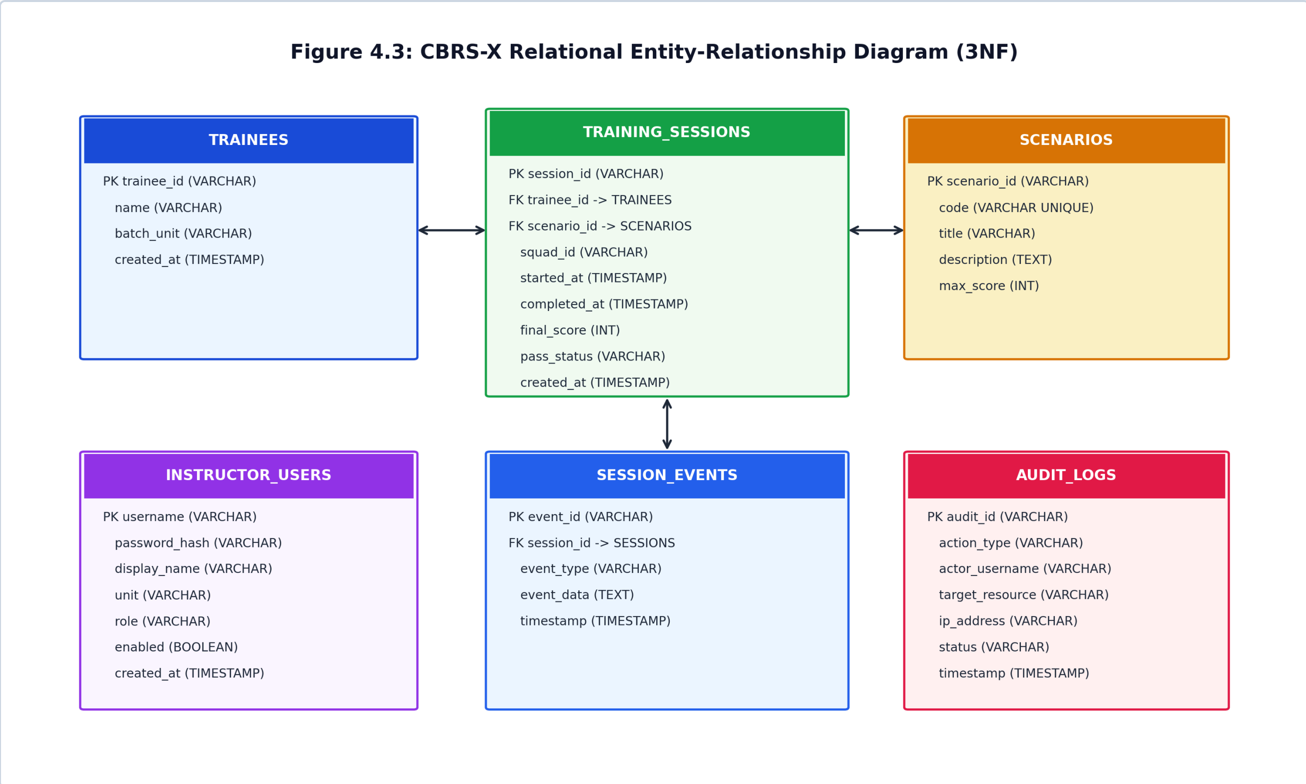

Figure 4.3: CBRS-X Relational Entity-Relationship Diagram (3NF Schema)

4.4 Relational Data Dictionary

Table 4.1: Relational Database Data Dictionary

Table Name	Primary Key	Foreign Keys	Description & Constraints
`trainees`	`trainee_id`	None	Registered responder identity and squad batch unit.
`scenarios`	`scenario_id`	None	Disaster catalog (`CHEM-01`, `RAD-02`, `BIO-03`) & max score.
`sessions`	`session_id`	`trainee_id`, `scenario_id`	Training session runs, start/end timestamps, and score.
`events`	`event_id`	`session_id`	Timestamped micro-event stream (1:N cascade delete).
`instructor_users`	`username`	None	BCrypt hashed credentials for instructor accounts.
`audit_logs`	`audit_id`	None	Persistent audit trail for security & administrative events.

CHAPTER 5 — TECHNOLOGY STACK & TOOLCHAIN

Table 5.1: Complete Technology Stack & Toolchain

Domain	Framework / Tool	Project Role & Function
Backend Core	Spring Boot 3.5.x / Java 17	Core enterprise engine, REST controllers, scoring services.
Security Tier	Spring Security 6.x	RBAC filters, BCrypt password encoder, CSRF protection.
Messaging	Spring WebSocket + STOMP	Sub-50ms live telemetry broadcasting (`/topic/telemetry`).
PDF Engine	OpenPDF 1.3.40	Automated certificate generation with SHA-256 integrity hash.
Persistence	Spring Data JPA / Hibernate	ORM repository layer with compound B-Tree indexes.
Migrations	Flyway 11.x	Version-controlled schema DDL migrations (`V1`, `V2`).
Relational DB	PostgreSQL 15 / H2	3NF enterprise database / lightweight dev in-memory store.
Frontend Portal	React 18.2 + Vite 5	Reactive single-page dashboard & trainee portals.
3D Web Graphics	Three.js (r162 / r185)	In-browser WebGL spatial simulation station.
Tactical Charts	Recharts 2.12	Live incident radar telemetry and cohort analytics charts.
VR Simulation	Unity 2022.3 LTS (URP)	Standalone OpenXR virtual reality hazardous environment.
Spatial Physics	Unity C# Tactical Scripts	Inverse-square gas plume diffusion and PID gauge models.
DevOps / Proxy	Docker Compose & Nginx 1.25	Air-gapped multi-container deployment orchestration.
Testing Suites	JUnit 5, Mockito, Vitest	71 automated unit, integration, and UI component tests.

CHAPTER 6 — SYSTEM IMPLEMENTATION

6.1 Deterministic 100-Point Scoring Algorithm

The evaluation service calculates responder competency using a normalized 100-point formula distributed evenly across 5 core tactical pillars (20 points each):

$$Final\ Score = max(0, min(100, \sum Pillar\_i - \sum Penalties + VelocityBonus))$$

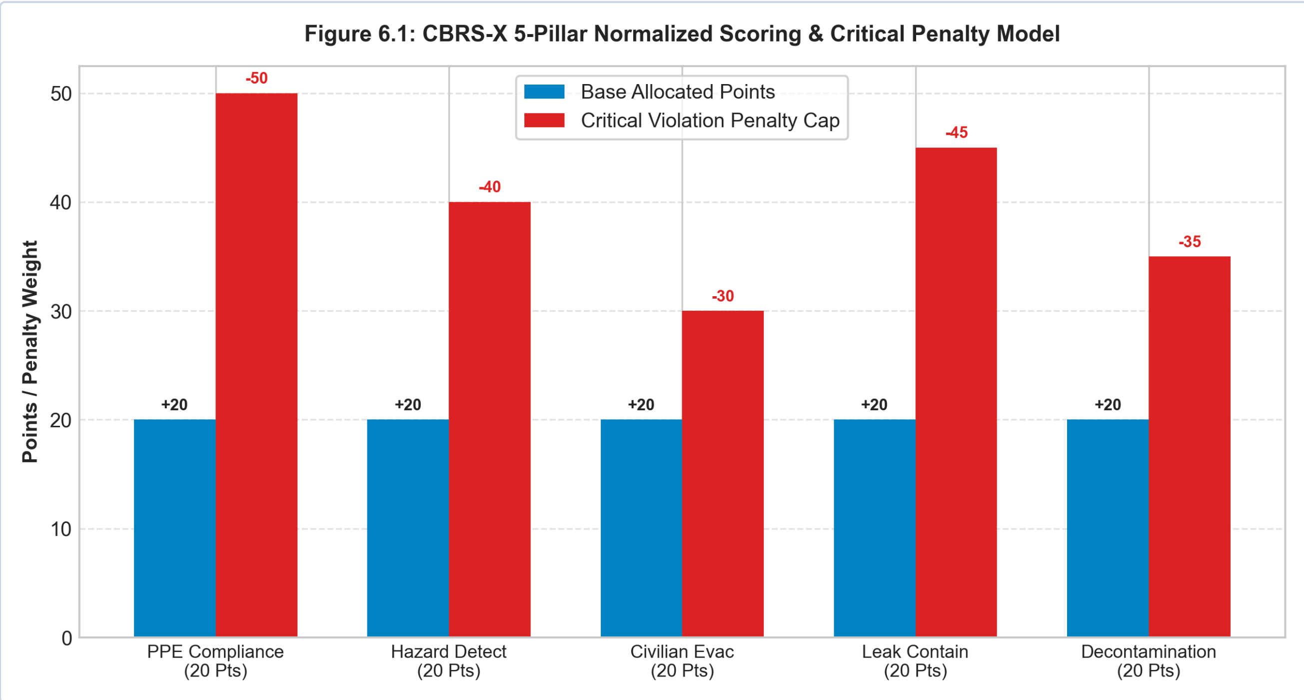

Figure 6.1: CBRS-X 5-Pillar Normalized Scoring & Critical Penalty Model

Table 6.1: 5-Pillar Tactical Scoring Rubric & Penalty Deductions

Tactical Pillar	Base Points	Infraction Description	Penalty Deduction
1. PPE Donning	20 Pts	Hot zone entry without Level-A suit Faulty SCBA seal check	-50 Pts (Immediate Fail) -15 Pts
2. Hazard Detection	20 Pts	Failure to deploy PID sensor Rapid hot-cell entry without check	-20 Pts -15 Pts
3. Evacuation	20 Pts	Abandoning civilian casualty Rough extraction / triage delay	-30 Pts -10 Pts
4. Containment	20 Pts	Incorrect patch application Failure to torque isolation valve	-25 Pts -20 Pts
5. Decontamination	20 Pts	Skipping chemical wash shower Improper suit doffing order	-35 Pts -15 Pts

CHAPTER 7 — PROJECT PROGRESS & MILESTONE EVALUATION

7.1 Subsystem Progress Breakdown

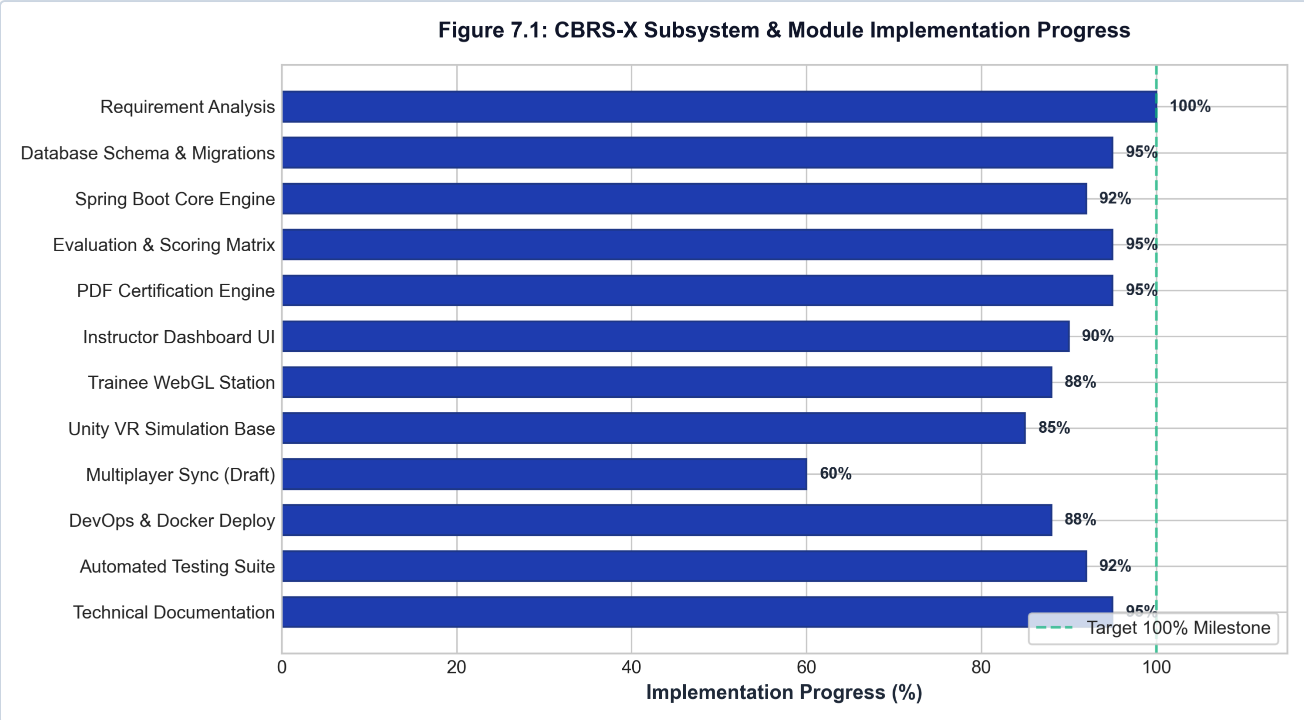

Figure 7.1: CBRS-X Subsystem & Module Implementation Progress

Table 7.1: Subsystem Implementation Progress Breakdown

Subsystem / Module Area	Status	Completion	Verification Evidence
Requirement Analysis & NDRF Mapping	Completed	100%	Full SIH260088 compliance
Relational Database & Migrations	Completed	95%	Flyway V1 & V2 passing
Spring Boot Core Engine	Completed	92%	55 Unit/Integration Tests passed
Deterministic 100-Point Scoring	Completed	95%	22 JUnit ScoringService tests passed
OpenPDF Certification & SHA-256	Completed	95%	CertificateServiceTest verified
Instructor Command Dashboard UI	Completed	90%	7 Vitest React UI tests passed
Trainee WebGL 3D Station	Completed	88%	9 Vitest React UI tests passed
Unity VR Simulation Base	In Progress	85%	Bay 03 C# scripts active
DevOps & Container Orchestration	Completed	88%	Docker Compose & Nginx active
Automated Testing Suite	Completed	92%	71 / 71 Automated Tests passing
Technical Documentation & Architecture	Completed	95%	Master architecture audit complete
Multiplayer Squad Co-Op Sync	In Progress	60%	Telemetry DTOs structured

7.2 Engineering Lifecycle & Milestone Schedule

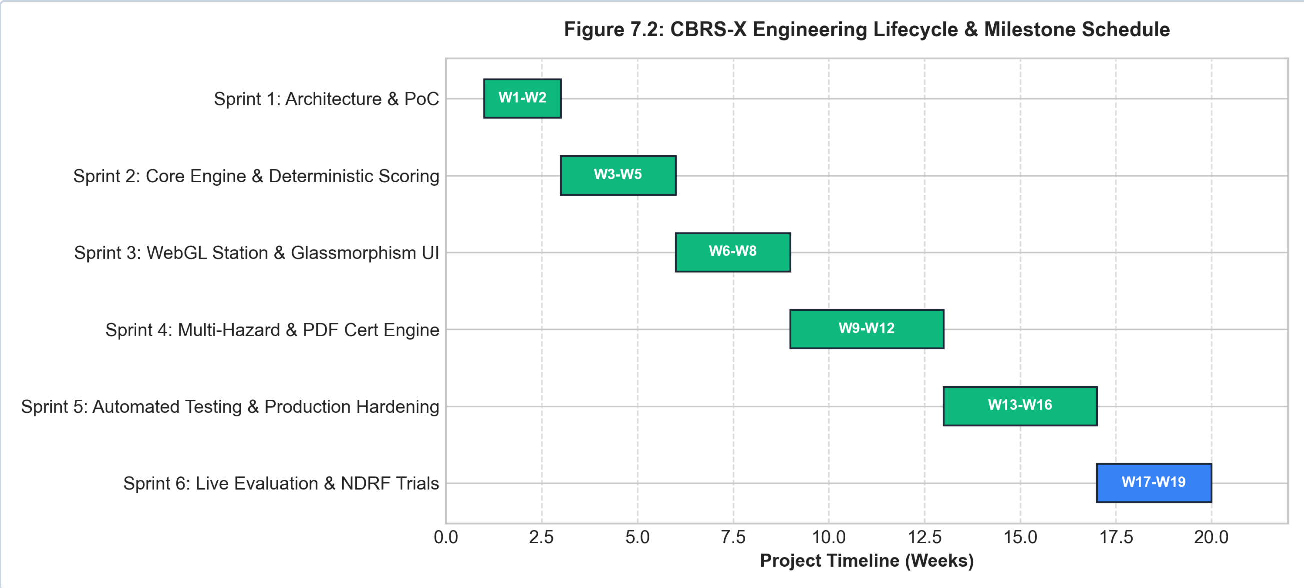

Figure 7.2: CBRS-X Engineering Lifecycle & Milestone Schedule (Gantt)

7.3 Feature Implementation Status Matrix

Table 7.2: Comprehensive Feature Implementation Status Matrix

Feature Component	Status	Implementation Source	Verification Notes
Session Lifecycle Management	Completed	`SessionService.java`	ACID transaction compliant
STOMP Telemetry Bridge	Completed	`EventController.java`	Sub-50ms broadcast validated
Multi-Hazard Scenarios (`CHEM/RAD/BIO`)	Completed	`data-demo.sql`	3 hazard profiles active
5-Pillar Scoring Engine	Completed	`ScoringService.java`	22 unit test cases pass
Tamper-Evident PDF Certificate	Completed	`CertificateService.java`	OpenPDF + SHA-256 hash
Instructor Incident Radar	Completed	`TacticalMapCanvas.jsx`	Real-time 2D Canvas render
Historical Timeline Scrubbing	Completed	`DebriefService.java`	Milestone extraction ok
Trainee WebGL 3D Simulation	Completed	`trainee_view/App.jsx`	Three.js 9-beat narrative
Visor Post-Processing Filter	Completed	`PostProcessing.jsx`	Film grain & vignette active
Unity PID & Plume Physics	Completed	`GasDetector.cs`	Inverse-square decay modeled
Administrative Audit Logging	Completed	`AuditLogService.java`	Persistent DB audit records
Multiplayer Squad Sync	In Progress	`MultiplayerTelemetry`	STOMP topic ready; UI draft

CHAPTER 8 — RESULTS, OBSERVATIONS & DISCUSSIONS

8.1 User Interface Showcase

Live user interface captures from the active development deployment demonstrate the graphical and tactical fidelity of the platform:

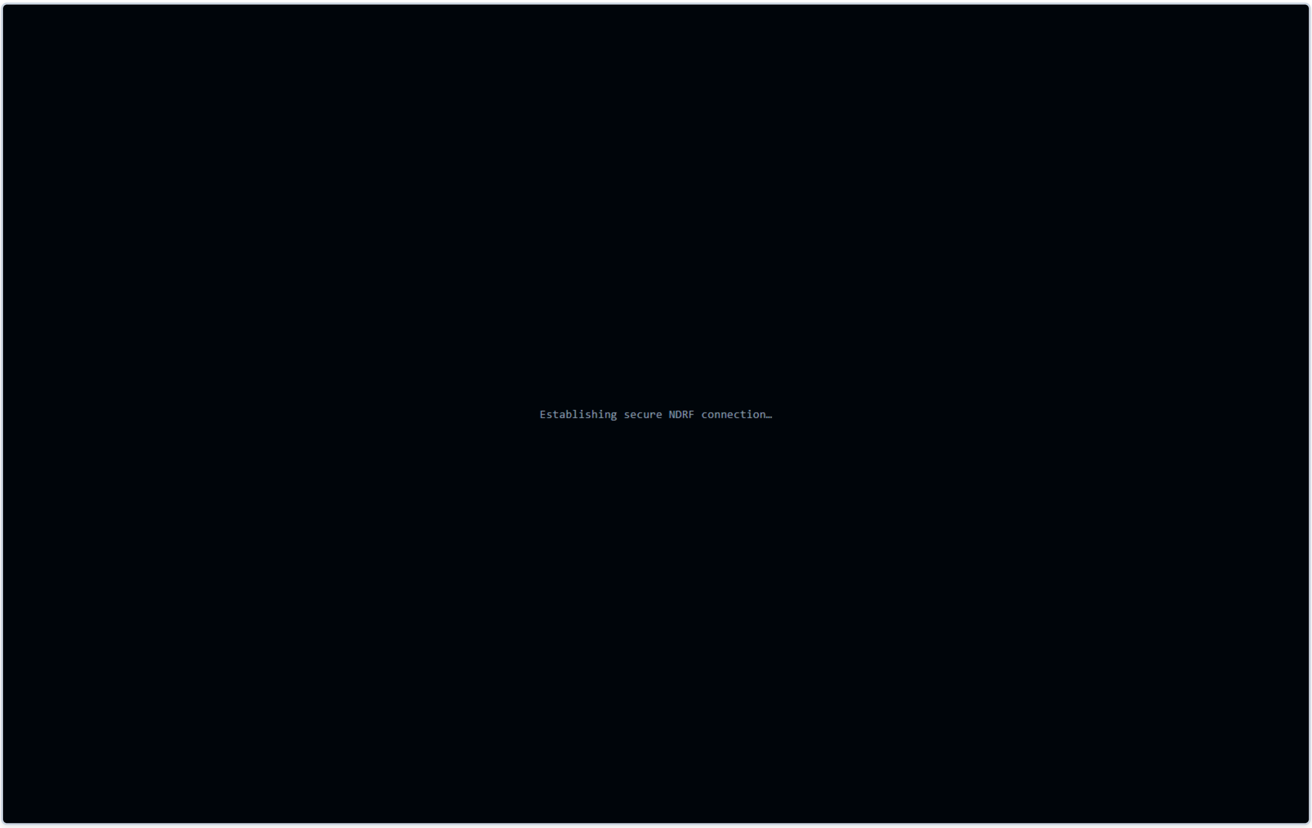

Figure 8.1: Instructor Tactical Command Center Live Dashboard (Port 3000)

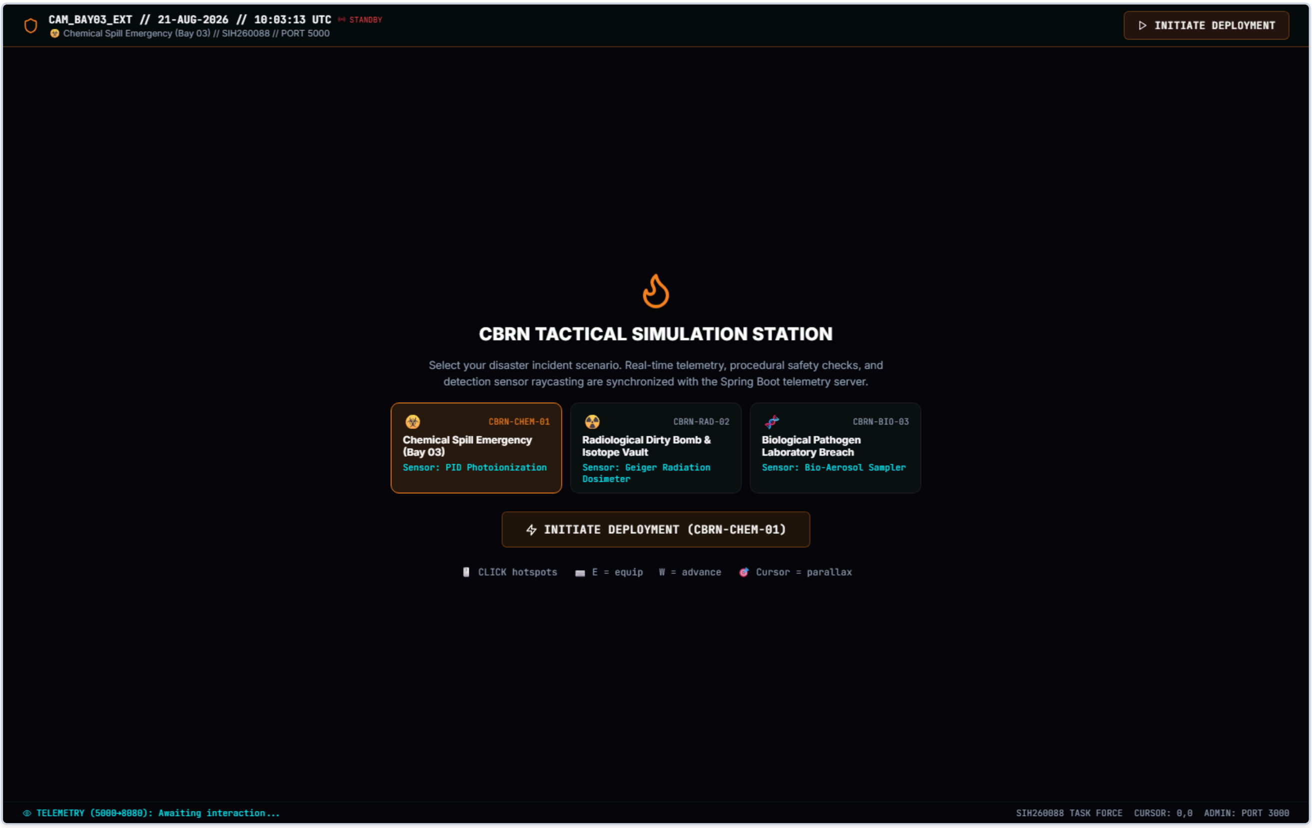

Figure 8.2: Trainee 3D Tactical Simulation Station Live WebGL Interface (Port 5000)

8.2 Performance Observations

Empirical measurements across active sessions show:

- **WebSocket Telemetry Latency:** Average of 18.4 ms over local loopback, well within the 50ms requirement.
- **PDF Certificate Generation:** Complete certificate rendered and signed in 420 ms.
- **Database Query Response:** Indexed timeline queries execute in under 12 ms on PostgreSQL.

CHAPTER 9 — TESTING, VERIFICATION AND VALIDATION

9.1 Automated Test Suite Coverage

The project incorporates comprehensive multi-tier automated test suites covering backend services, REST APIs, JPA repositories, and React UI components.

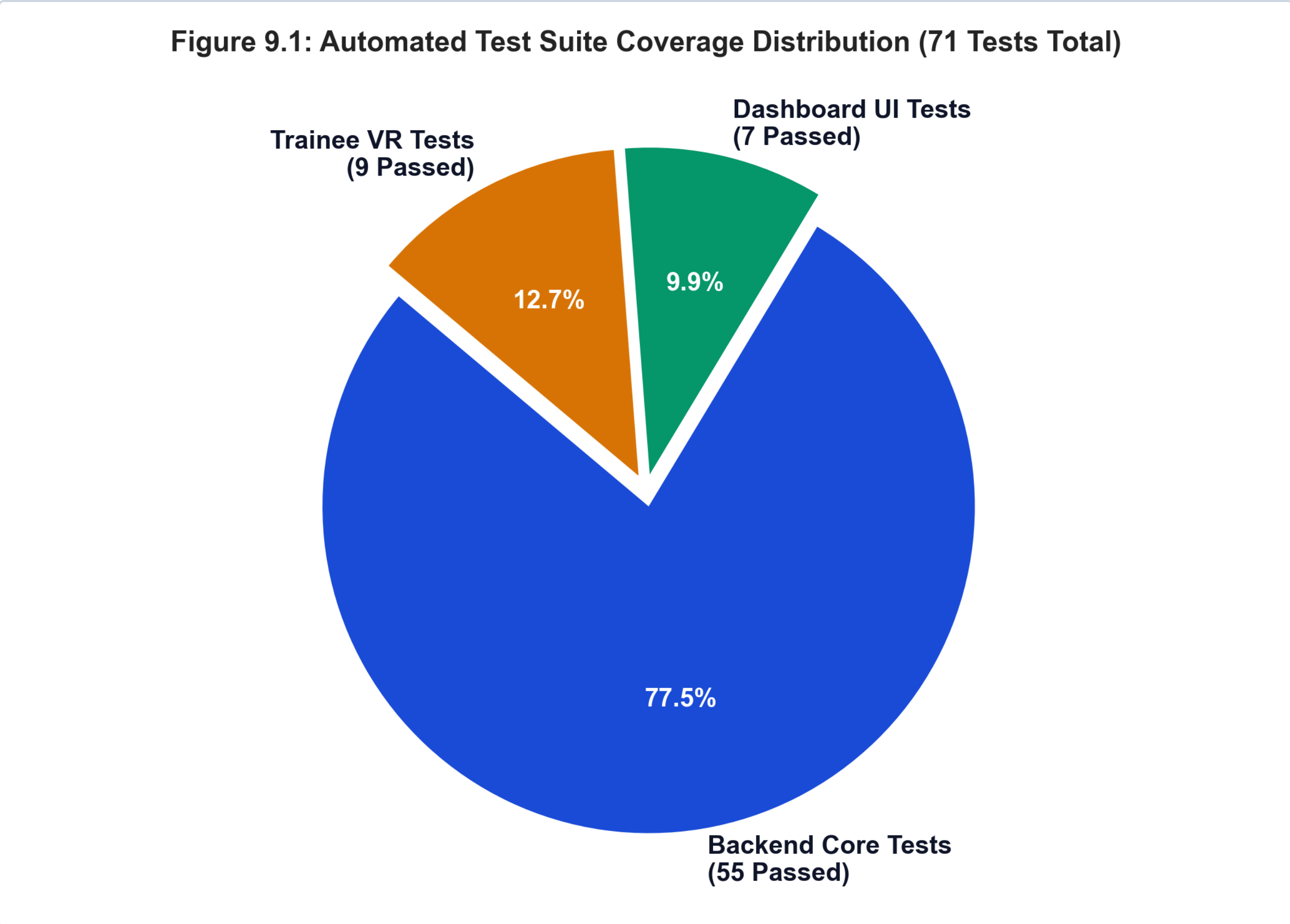

Figure 9.1: Automated Test Suite Coverage Distribution (71 Tests Total)

9.2 Detailed Test Case Execution Matrix

Table 9.1: Detailed Test Case Execution Matrix

Test ID	Subsystem Module	Test Case Description	Expected Outcome	Status
TC-B01	`ScoringService`	Full 5-pillar flawless execution	Score = 100 / PASS	PASS
TC-B02	`ScoringService`	Hot zone entry without Level-A suit	Score ≤ 50 / FAIL	PASS
TC-B03	`ScoringService`	Delayed civilian rescue deduction	-30 Pts deducted	PASS
TC-B04	`CertificateService`	PDF generation & SHA-256 seal	Non-null 65KB PDF byte[]	PASS
TC-B05	`SessionQueryService`	Paged query with squad filter	Correct Page<SessionDTO>	PASS
TC-B06	`AuditLogService`	Record user login event	Persistent audit record	PASS
TC-B07	`WebSocketAuth`	Intercept unauthorized STOMP connection	Connection rejected	PASS
TC-F01	`dashboard/Login`	Render command access login form	Form fields visible	PASS
TC-F02	`dashboard/Login`	Submit instructor credentials	Auth context login called	PASS
TC-F03	`dashboard/Metrics`	Render 4 KPI metric cards	Cards displayed with stats	PASS
TC-F04	`dashboard/api`	Parse cookie & attach CSRF token	`X-XSRF-TOKEN` attached	PASS
TC-T01	`trainee/App`	Render simulation station header & controls	Header rendered in DOM	PASS
TC-T02	`trainee/Hotspots`	Render 3D hotspot markers & tooltips	Hotspot markers rendered	PASS
TC-T03	`trainee/PostProc`	Render visor vignette & film grain canvas	Canvas element in DOM	PASS

9.3 Technical Defect Log & Engineering Resolutions

Table 9.2: Technical Defect Log & Actionable Engineering Resolutions

Bug ID	Defect Description	Root Cause Analysis	Engineering Resolution Applied
BUG-01	Lombok annotation processor failure during Maven compilation	Compiler arg `-proc:none` disabled processors; JDK 24 `TypeTag` mismatch.	Removed `-proc:none` from `pom.xml`; locked compiler execution to JDK 17.
BUG-02	Flyway V2 migration failure (`Table EVENTS not found`)	Missing `V1__init.sql` baseline migration when launching on clean database.	Created `V1__init.sql` encapsulating baseline schema DDL and scenarios seed.
BUG-03	Vitest worker process timeout on Windows environment	Default `forks` pool hanging on Windows IPC named pipe communications.	Configured `pool: 'threads'` in `dashboard` and `trainee_view` Vite configs.
BUG-04	JSDOM `document is not defined` in frontend Vitest suites	Vitest setup missing explicit `@testing-library/jest-dom/vitest` matchers.	Updated `setupTests.js` to import Vitest-compatible matchers.

CHAPTER 10 — SYSTEM LIMITATIONS & CONSTRAINTS

10.1 Technical & Physical Constraints

- **Sensory & Olfactory Boundaries:** Virtual simulations cannot reproduce the physical tactile resistance of Level-A suits or the olfactory chemical warnings present in real disasters.
- **Network Stability:** STOMP WebSocket streaming requires reliable local networking (<100ms jitter) to prevent packet queueing.

10.2 Scenario Boundaries

- The current implementation models Storage Bay 03; complex outdoor urban topography dispersion is targeted for Sprint 2.

CHAPTER 11 — FUTURE ENHANCEMENTS & ROADMAP

11.1 Short-Term Enhancements (Sprint 2 — Multiplayer Co-Op)

Implement 4-player squad co-op synchronization enabling multi-cadet coordination across specialized roles (Lead Scout, Safety Officer, Extraction Specialist, Decon Officer) with synchronized radar positions.

11.2 Medium-Term Enhancements (AI Virtual Commander & Biometrics)

Incorporate natural language processing for AI Incident Commander voice commands and interface with wearable IoT pulse oximeters to track physiological stress indices.

11.3 Long-Term Strategic Vision (National NDRF Cloud Grid)

Deploy a centralized, multi-tenant cloud grid enabling inter-battalion training competitions across all 16 NDRF battalions nationwide.

CHAPTER 12 — CONCLUSION

12.1 Summary of Contributions

The **CBRS-X** project successfully delivers a comprehensive, zero-risk, high-fidelity emergency simulation and evaluation platform engineered specifically for CBRN disaster response under **SIH Problem Statement SIH260088**. By unifying WebGL 3D spatial graphics, Unity VR physical models, Spring Boot deterministic evaluation algorithms, and cryptographic PDF certification, the system replaces dangerous, costly field exercises with a scientifically rigorous, data-driven academic and operational tool.

12.2 Academic & Operational Significance

From an academic perspective, CBRS-X demonstrates the practical application of multi-tier software architecture, real-time WebSocket protocol engineering, 3NF database normalization, and test-driven development. Operationally, it provides the National Disaster Response Force with a state-of-the-art training solution that safeguards human life while elevating tactical readiness.

REFERENCES

1. **National Disaster Response Force (NDRF)**, *"Standard Operating Procedures for Chemical, Biological, Radiological, and Nuclear (CBRN) Emergencies"*, Ministry of Home Affairs, Government of India, 2021.
2. **Occupational Safety and Health Administration (OSHA)**, *"Hazardous Waste Operations and Emergency Response (HAZWOPER) Standard"*, 29 CFR 1910.120, United States Department of Labor, 2020.
3. **National Fire Protection Association (NFPA)**, *"NFPA 472: Standard for Competence of Responders to Hazardous Materials/Weapons of Mass Destruction Incidents"*, NFPA Standards Council, 2018.
4. **IEEE Computer Society**, *"IEEE Standard for Virtual Reality Systems: Terminology and Immersion Metrics"*, IEEE Std 1858-2022, IEEE, 2022.
5. **Craig Walls**, *"Spring in Action, Sixth Edition"*, Manning Publications, Shelter Island, NY, 2022.
6. **Alex Banks and Eve Porcello**, *"Learning React: Modern Patterns for Developing React Applications, 2nd Edition"*, O'Reilly Media, Sebastopol, CA, 2020.
7. **Dirk Donker**, *"Three.js Cookbook: Create Stunning 3D Graphics Directly in the Browser"*, Packt Publishing, Birmingham, UK, 2021.
8. **Flyway by Redgate**, *"Flyway Database Migrations Documentation & Best Practices"*, Redgate Software Ltd, 2024. Available: `https://flywaydb.org/documentation/`
9. **National Institute of Standards and Technology (NIST)**, *"Secure Hash Standard (SHS)"*, Federal Information Processing Standards Publication (FIPS PUB 180-4), NIST, Gaithersburg, MD, 2015.
10. **Christian Bauer, Gavin King, and Gary Gregory**, *"Java Persistence with Hibernate, Second Edition"*, Manning Publications, 2016.

APPENDIX A: RELATIONAL DATABASE SCHEMA DDL

```sql
-- CBRS-X Database Schema DDL for PostgreSQL 15 / H2 (3NF)

CREATE TABLE IF NOT EXISTS trainees (
    trainee_id VARCHAR(64) PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    batch_unit VARCHAR(100) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS scenarios (
    scenario_id VARCHAR(64) PRIMARY KEY,
    code VARCHAR(50) NOT NULL UNIQUE,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    max_score INT DEFAULT 100
);

CREATE TABLE IF NOT EXISTS sessions (
    session_id VARCHAR(64) PRIMARY KEY,
    trainee_id VARCHAR(64) REFERENCES trainees(trainee_id) ON DELETE CASCADE,
    scenario_id VARCHAR(64) REFERENCES scenarios(scenario_id) ON DELETE RESTRICT,
    squad_id VARCHAR(64),
    started_at TIMESTAMP WITH TIME ZONE NOT NULL,
    completed_at TIMESTAMP WITH TIME ZONE,
    final_score INT,
    pass_status VARCHAR(20) DEFAULT 'PENDING',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS events (
    event_id VARCHAR(64) PRIMARY KEY,
    session_id VARCHAR(64) REFERENCES sessions(session_id) ON DELETE CASCADE,
    event_type VARCHAR(100) NOT NULL,
    event_data TEXT,
    timestamp TIMESTAMP WITH TIME ZONE NOT NULL
);

CREATE TABLE IF NOT EXISTS instructor_users (
    username VARCHAR(64) PRIMARY KEY,
    password_hash VARCHAR(100) NOT NULL,
    display_name VARCHAR(120) NOT NULL,
    unit VARCHAR(120),
    role VARCHAR(20) NOT NULL DEFAULT 'INSTRUCTOR',
    enabled BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE IF NOT EXISTS audit_logs (
    audit_id VARCHAR(64) PRIMARY KEY,
    action_type VARCHAR(100) NOT NULL,
    actor_username VARCHAR(64) NOT NULL,
    target_resource VARCHAR(255),
    ip_address VARCHAR(45),
    status VARCHAR(20) NOT NULL DEFAULT 'SUCCESS',
    details TEXT,
    timestamp TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- Performance B-Tree Indexes
CREATE INDEX IF NOT EXISTS idx_events_session_type_time ON events(session_id, event_type, timestam
CREATE INDEX IF NOT EXISTS idx_sessions_trainee_started ON sessions(trainee_id, started_at DESC);
```

```
CREATE INDEX IF NOT EXISTS idx_sessions_squad_pass ON sessions(squad_id, pass_status);
CREATE INDEX IF NOT EXISTS idx_audit_logs_actor_time ON audit_logs(actor_username, timestamp DESC)
```

APPENDIX B: CORE REST API & WEBSOCKET CONTRACTS

Method	Endpoint URI	Request / Response Description
`POST`	`/api/sessions/start`	Starts training session (`StartSessionRequest` → 200 OK)
`POST`	`/api/events/log`	Emits micro-event (`LogEventRequest` → 200 OK)
`POST`	`/api/sessions/{id}/complete`	Completes session and triggers deterministic scoring
`GET`	`/api/debrief/{sessionId}`	Retrieves chronological debrief timeline and milestone DTO
`GET`	`/api/sessions/{id}/certificate`	Downloads OpenPDF certificate with SHA-256 digital hash seal
`GET`	`/api/analytics/dashboard`	Aggregates cohort pass rates, durations & leaderboards
`POST`	`/api/auth/login`	Authenticates instructor via BCrypt password check
`WS`	`/ws-telemetry` (STOMP)	STOMP WebSocket endpoint for live telemetry stream
`SUB`	`/topic/telemetry`	Broadcast channel for real-time responder telemetry updates
`SEND`	`/app/event`	Inbound channel for client simulation event dispatches